City Map Architecture Overview

The City Map service stores the capability topology as a directed graph. Nodes are capabilities; edges are typed relationships (depends_on, registered_in). The graph is persisted in DynamoDB and exposed through a GraphQL API consumed by the Capability Registry and deploy pipelines.

Write path: ownership changes enter through the City Map UI, are validated (owner_team required, approver on tier-1 changes), and published to the capability-events Kafka topic.

Data Flow and Consumers

The Capability Registry consumes capability-events and syncs ownership every 15 minutes. Deploy pipelines query the City Map at release time to block deploys of capabilities whose dependencies are not Active.

Disaster recovery: the graph is snapshotted to S3 hourly. Restoring from snapshot takes under 10 minutes and is drilled quarterly by the platform team (owner: jordan.lee).